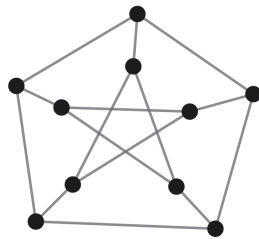
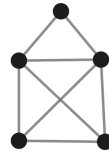
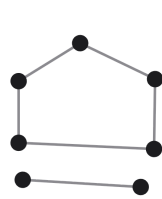
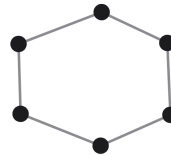


1. Determine which of the following graphs have spanning trees.



Petersen Graph



**Solution:** The disconnected one does not have a spanning tree, but the connected ones do.

2. For the graphs above that do have spanning trees, find a spanning tree for each of them and 2-color it.

**Solution:** Exercise

3. Let  $\mathcal{C}$  be the set of all finite binary sequences, including the empty sequence  $()$ . Let  $S \subseteq \mathcal{C}$ . Consider the graph  $G = (V, E)$  where  $V = S$  and for any two sequences  $s, t$ , we have  $\{s, t\} \in E$  iff either  $s$  extends  $t$  by one bit or  $t$  extends  $s$  by one bit. So for example, there is an edge between 1001 and 10010 but not between 1001 and 01010 (the empty string  $()$  is just adjacent to 0 or 1, if either of them are in  $S$ ).

Prove that if  $S$  contains  $()$  and is closed downward (i.e. if  $t \in S$  and  $s$  is the first  $n$  bits of  $t$  for some  $n$  less than the length of  $t$ , then  $s \in S$ ), then  $G$  is a tree.

**Solution:** First, note that  $G$  has no cycles. Otherwise, suppose there is a cycle that begins and ends with  $s$ . WLOG assume  $s$  is the shortest string in its cycle. Since  $s$  must be adjacent to at least two other strings in this cycle, they must be  $s \frown 0$  and  $s \frown 1$ . But then, there is a path between  $s \frown 0$  and  $s \frown 1$  that does not contain  $s$ , which is a contradiction.

To see that  $G$  is connected, it suffices to show that there is a path between every  $s \in S$  and  $()$ . To reach  $()$ , just take  $s$  and keep chopping off bits until you reach  $()$  (this is where we use downward closure).

4. Using the definitions from problem 3, prove that if  $S$  contains  $()$ , then  $G$  is a tree iff  $S$  is closed downward. (you just proved one of the directions)

**Solution:** It suffices to show that if  $G$  is connected, then  $S$  is closed downward. Let  $s \in S$ , where  $s = b_1 b_2 \dots b_n$ . Since  $() \in S$ , we know there is a path connecting  $()$  and  $s$ . But the only such path is  $b_1, b_1 b_2, b_1 b_2 b_3$ , etc., which contains every substring of  $s$ .